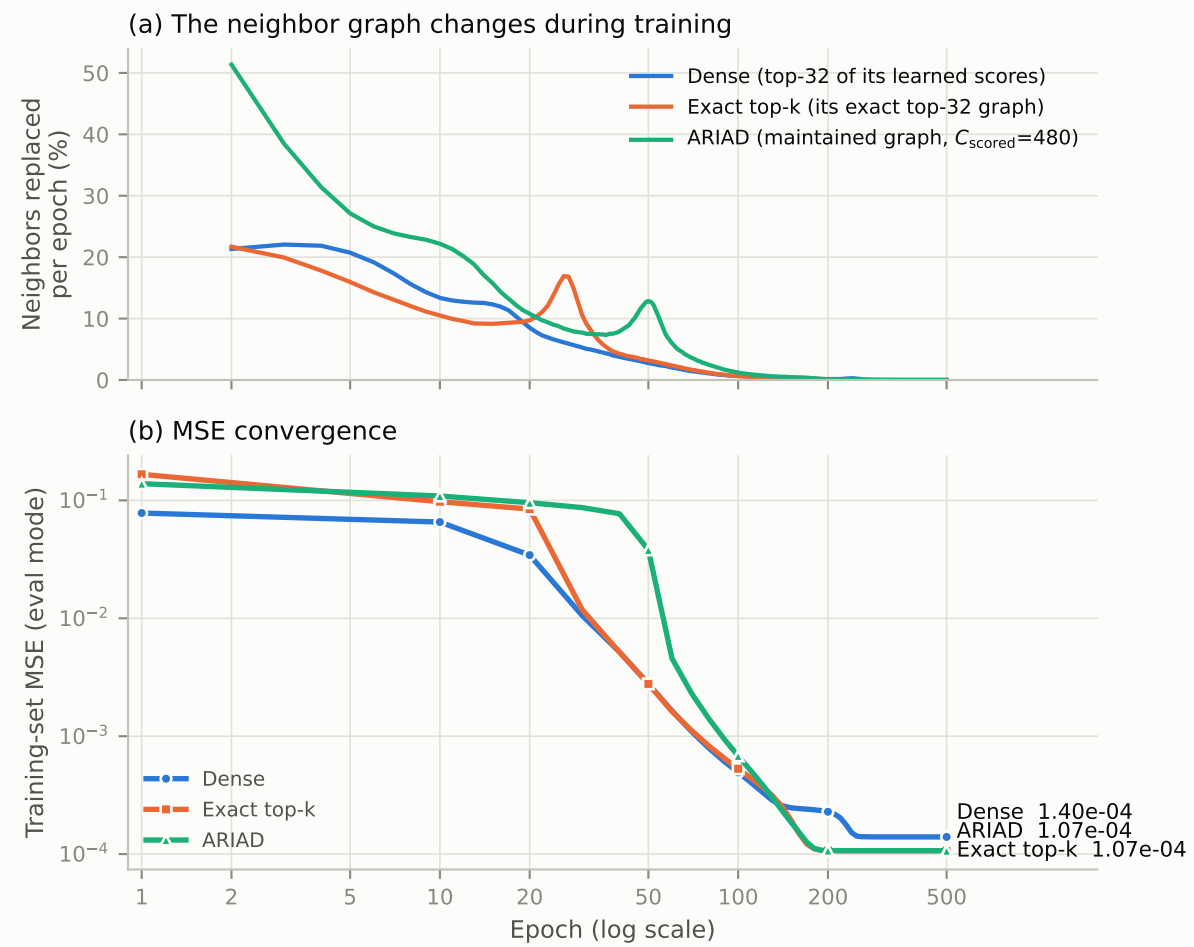


Training dynamics (N=10,000, k=32, sparse top-32 teacher)



(a) Each method's own 32-neighbor graph H_t in its own run, audited on all N rows every epoch (not part of training cost); replacement = $1 - |H_t \cap H_{t-1}| / 32$, averaged over rows. Dense attends to all nodes; its line tracks the top-32 of its learned scores. ARIAD's early peak is its first refreshes away from the random initial graph.
(b) MSE at epoch 1 and every 10 epochs. Both panels come from the same seed-0 training runs (full batch, 1 step = 1 epoch, Adam $\text{lr}=1\text{e-}2$, identical initialisation for all methods; untied W_q/W_k , fixed values V).